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Equation 1

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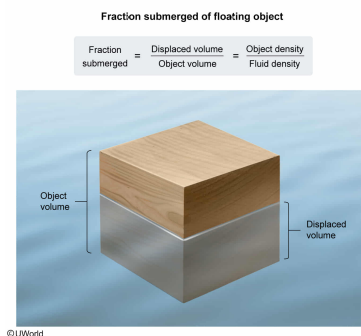
A 0.3-kg object with half the density of human fatty tissue is placed in water. Ignoring the atmospheric pressure, what percentage of the mass will be submerged?

- ☐ A. 30% [15%]
- ✓ ☒ B. 45% [74%]
- ☐ C. 90% [6%]
- ☐ D. 100% [3%]

Correct

75% answered correctly

Explanation:



According to the [Archimedes principle](#), an object in a fluid experiences an upward buoyant force equal to the weight of the displaced fluid. If the object's density is less than the fluid's density, the object will [float at the surface](#) because the buoyant force is in equilibrium with the weight of the object. For a floating object, the **fraction submerged** is equal to the ratio of the object's density to the fluid's density:

$$\text{fraction submerged} = \frac{\rho_{\text{object}}}{\rho_{\text{fluid}}}$$

The density of human fat is given in the passage as 0.9 kg/L, and the **object's density** is half this value; ρ_{object} is **0.45 kg/L**. Because the object is less dense than water (ρ_{water} is 1 kg/L), the object will float at the surface and be partially submerged. Using these values, the fraction submerged is:

$$\frac{0.45 \text{ kg/L}}{1 \text{ kg/L}} = 0.45$$

Therefore, **45% of the object** will be submerged.

(Choice A) The fraction submerged depends on the density of the object, not its mass.

(Choice C) If the density of fat is used as the density of the object, the incorrectly calculated percent submerged would be twice as large, or 90%.

(Choice D) If the density of the object is greater than or equal to the density of the fluid, the object will be fully (100%) submerged. The given object is less dense than water and therefore will be only partially submerged.

Educational objective:

An object floats at the surface of a fluid if its density is less than that of the fluid. The fraction submerged is equal to the ratio of the object's density to the fluid density.

Time spent:0 sec
Subject:
Physics

QID:401071
System:
Fluids

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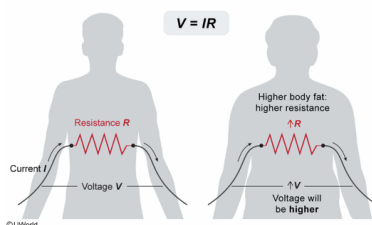
Fatty tissue has a lower electrical conductivity than lean (non-fat) tissue. This property allows body fat content to be estimated by modeling the tissue between two points as a single resistor. For a constant current entering the body, relatively high body fat content would be indicated by:

- ☐ A. an increased current leaving the body. [2%]
- ☐ B. a decreased current leaving the body. [37%]
- ☐ C. a lower voltage difference across the body. [20%]
- ☒ D. a higher voltage difference across the body. [39%]

Correct

38% answered correctly

Explanation:



Electrical conductivity is the measure of how easily a current can move through a material, and it is inversely proportional to electrical resistance. The question states that fatty tissue is less conductive than lean tissue; in other words, fatty tissue is more resistive. Therefore, an individual with a relatively **high body fat** content will have a relatively **high resistance**.

The question also states that the current entering the body is constant. Current is the movement of electric charges, and **electric charge** is always **conserved**; it cannot be created or destroyed. Because current cannot be gained (increased) or lost (decreased) across a resistor, the amount of **current** entering and exiting a resistor is always the **same**.

Voltage is the difference in **electric potential** between two points that drives the movement of electric charges. **Ohm law** describes the **voltage difference** V across a resistor, which is the product of the current I and the resistance R :

$$V = IR$$

Therefore, the voltage difference measured across the resistor is proportional to the resistance, and a high body fat content (**high resistance**) in an individual will be indicated by a **high voltage** difference between two points.

(Choices A and B) Conservation of electric charge implies that the amount of current exiting the body must be constant because the current entering the body is constant.

(Choice C) A relatively high body fat content increases the resistance of the body. When current is constant, higher resistance will increase the voltage difference across the body, not lower the voltage difference.

Educational objective:

The current entering and exiting a resistor remains the same because electric charge is always conserved. The voltage across a resistor is the product of resistance and current ($V = IR$). Therefore, tissues with a higher resistance will have a higher voltage across them for the same amount of current.

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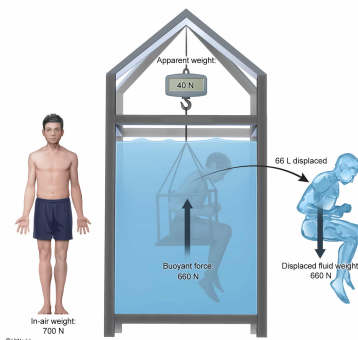
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An individual that weighs 700 N in air has an apparent weight of 40 N when submerged in water. What is the volume of the displaced water?

- ☐ A. 40 L [17%]
- ☐ B. 44 L [13%]
- ✓ ☒ C. 66 L [58%]
- ☐ D. 70 L [11%]

Correct
58% answered correctly

Explanation:



According to Archimedes principle, a submerged object experiences an upward buoyant force F_B due to the displaced fluid. Therefore, an object will appear to weigh less than its in-air weight W_{air} when submerged in water, given by the object's **apparent weight** W_{apparent} :

$$W_{\text{apparent}} = W_{\text{air}} - F_B$$

The F_B acting on the individual is the difference between W_{air} and W_{apparent} :

$$F_B = W_{\text{air}} - W_{\text{apparent}}$$

$$F_B = 700 \text{ N} - 40 \text{ N} = 660 \text{ N}$$

From [Archimedes principle](#), the **buoyant force** experienced by the individual is **equal to the weight of the displaced fluid**, which is the product of the density of water ρ , the displaced volume V , and the gravitational acceleration g :

$$F_B = \rho V g$$

The volume V displaced by the individual is calculated using the values of ρ_{water} and g given in the passage:

$$V = \frac{F_B}{\rho g}$$

$$V = \frac{660 \text{ N}}{(1 \text{ kg/L})(10 \text{ m/s}^2)} = 66 \text{ L}$$

The volume of the displaced water is **66 L**.

(Choice A) The apparent weight of the individual (40 N) is not equal to the buoyant force; the individual's apparent weight is the difference between the individual's in-air weight and the buoyant force.

(Choice B) Displacing 44 L of water would result in a buoyant force of 440 N and cause the individual's apparent weight to be 260 N instead of 40 N.

(Choice D) An amount of 70 L of displaced water would result in a buoyant force of 700 N. The buoyant force is equal to the in-air weight of the individual only if the individual is **floating at the surface** or if the individual was suspended (neither rising nor sinking).

Educational objective:

The apparent weight of a submerged object is its reduced weight due to an upward buoyant force. The buoyant force is equal to the weight of the displaced fluid, which is the product of the fluid's density ρ , volume of displaced fluid V , and the gravitational acceleration g : $F_B = \rho Vg$.

Time spent:0 sec

Subject:
Physics

QID:401073

System:
Fluids

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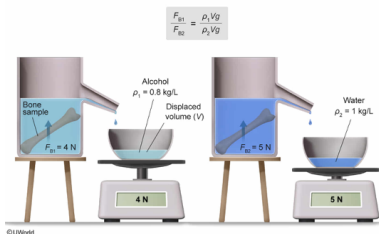
A bone sample (specific gravity = 3) completely immersed in isopropyl alcohol (specific gravity = 0.8) is subject to a buoyant force of 4 N. What buoyant force would the same bone sample experience when immersed in water?

- ☐ A. 2.4 N [17%]
☐ B. 4.0 N [12%]
☒ C. 5.0 N [62%]
☐ D. 12 N [8%]

Correct

62% answered correctly

Explanation:



The **specific gravity** of a liquid or solid substance is the ratio of its density to the density of water:

$$\text{specific gravity} = \frac{\rho_{\text{substance}}}{\rho_{\text{water}}}$$

The density of isopropyl alcohol ρ_{alcohol} is calculated from the specific gravity of isopropyl alcohol given in the question and the value of ρ_{water} given in the passage:

$$\rho_{\text{alcohol}} = (\text{specific gravity of alcohol})(\rho_{\text{water}})$$

$$\rho_{\text{alcohol}} = (0.8)(1 \text{ kg/L}) = 0.8 \text{ kg/L}$$

According to [Archimedes principle](#), a submerged object experiences an upward **buoyant force** F_B **equal to the weight of the displaced fluid**, which is the product of the fluid density ρ , the displaced volume V , and the gravitational acceleration g :

$$F_B = \rho V g$$

For an object **fully immersed** in two different fluids (1 and 2), the volumes of fluid displaced in each are equal to the volume of the object: $V_1 = V_2 = V_{\text{object}}$. Therefore, the **ratio of buoyant forces** is **equal to the ratio of fluid densities**:

$$\frac{F_{B,1}}{F_{B,2}} = \frac{\rho_1 V_1 g}{\rho_2 V_2 g} = \frac{\rho_1}{\rho_2}$$

Because the bone sample is fully immersed in isopropyl alcohol and water, the above relationship applies:

$$\frac{F_{B,\text{alcohol}}}{F_{B,\text{water}}} = \frac{\rho_{\text{alcohol}}}{\rho_{\text{water}}}$$

The buoyant force in water $F_{B,\text{water}}$ is calculated using the value of $F_{B,\text{alcohol}}$ given in the question, 0.8 kg/L for ρ_{alcohol} , and the value of ρ_{water} given in the passage:

$$F_{\text{B, water}} = \frac{\rho_{\text{water}}}{\rho_{\text{alcohol}}} (F_{\text{B, alcohol}})$$

$$F_{\text{B, water}} = \frac{1 \text{ kg/L}}{0.8 \text{ kg/L}} (4 \text{ N}) = 5 \text{ N}$$

The buoyant force experienced by the bone in water is **5 N**.

(Choices A and B) Because the buoyant force is proportional to the density of the fluid, and water is denser than isopropyl alcohol, the buoyant force in water must be greater than the buoyant force in isopropyl alcohol.

(Choice D) The buoyant force acting on a fully immersed object is independent of the density of the object.

Educational objective:

The magnitude of the buoyant force exerted on a submerged object is the product of the fluid's density ρ , the volume displaced V , and the gravitational acceleration g : $F_{\text{B}} = \rho V g$. For an object fully immersed in two different fluids, the ratio of the buoyant forces is equal to the ratio of the fluid densities.

Time spent: 0 sec

Subject:
Physics

QID: 401074

System:
Fluids

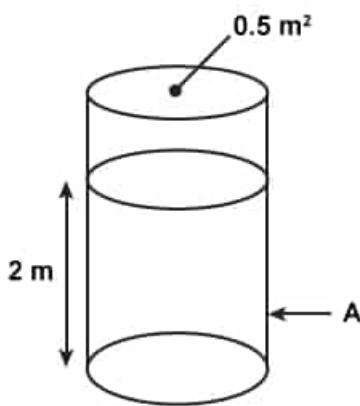
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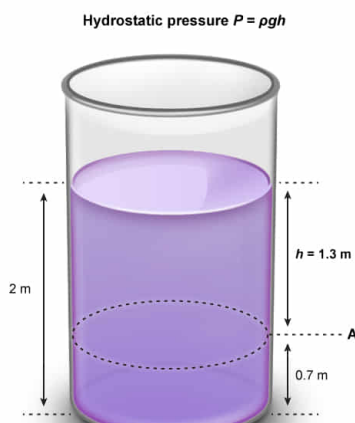
The hydrostatic weighing container above is filled with water to a depth of 2 m. If point A is 0.7 m above the bottom of the container, what is the pressure due to the weight of the water at point A? (Note: The density of water is 1,000 kg/m³.)

- ☐ A. $7.0 \times 10^3 \text{ N/m}^2$ [22%]
- ☐ B. $1.0 \times 10^4 \text{ N/m}^2$ [7%]
- ☒ C. $1.3 \times 10^4 \text{ N/m}^2$ [62%]
- ☐ D. $2.0 \times 10^4 \text{ N/m}^2$ [7%]

Correct

63% answered correctly

Explanation:



Hydrostatic pressure P is the **pressure exerted by the weight** of a static (nonmoving) fluid. It is typically measured relative to atmospheric pressure at sea level, also called gauge pressure. The magnitude of P is the product of the fluid density ρ , the gravitational acceleration g , and the height h of the fluid above the point of interest:

$$P = \rho gh$$

From the given figure, the **height** of the fluid **above point A** is the difference between the total height and the height of point A given in the question:

$$h = 2 \text{ m} - 0.7 \text{ m} = 1.3 \text{ m}$$

Hydrostatic pressure is calculated using the value of ρ_{water} given in the question, the value of g given in the passage, and the height h of 1.3 m:

$$P = \left(1,000 \frac{\text{kg}}{\text{m}^3}\right) \left(10 \frac{\text{m}}{\text{s}^2}\right) (1.3 \text{ m}) = 13,000 \frac{\text{N}}{\text{m}^2}$$

Converting to scientific notation, the hydrostatic pressure at point A is **$1.3 \times 10^4 \text{ N/m}^2$** .

(Choice A) The hydrostatic pressure exerted by the height of the fluid *below* point A ($h = 0.7 \text{ m}$) is $7.0 \times 10^3 \text{ N/m}^2$.

(Choice B) The *weight* (not the pressure) of all the fluid inside the container is calculated by $W = \rho Vg = 1.0 \times 10^4 \text{ N}$.

(Choice D) The hydrostatic pressure measured at the bottom of the container (the pressure exerted by the full height of the fluid, 2.0 m) is $2.0 \times 10^4 \text{ N/m}^2$.

Educational objective:

Hydrostatic pressure is the pressure exerted by the weight of a static fluid. It is the product of the fluid density ρ , the gravitational acceleration g , and the height of the fluid above a given point h : $P = \rho gh$.

Time spent: 0 sec

QID: 401075

Subject:
Physics

System:
Fluids

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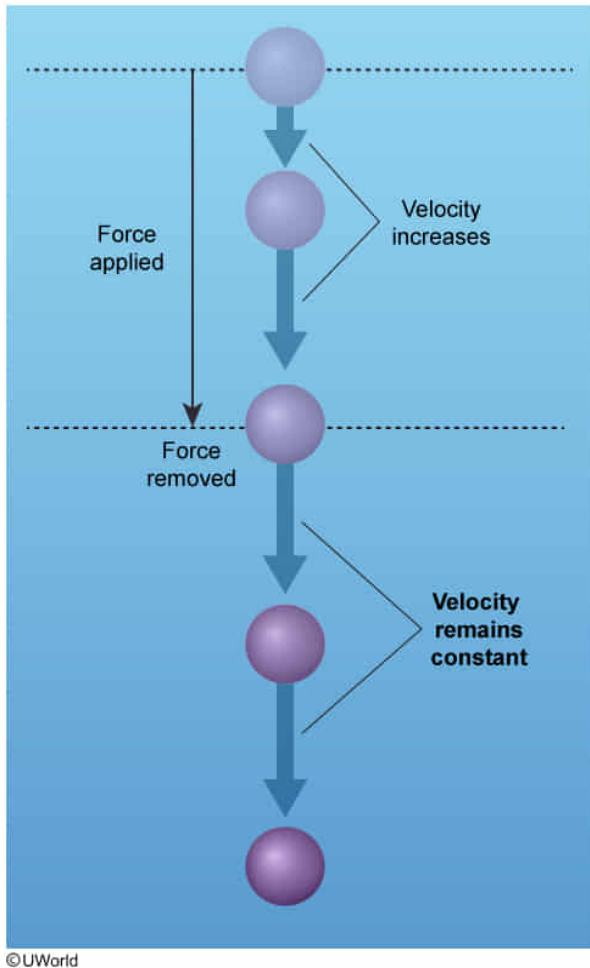
An object with the same density as water is stationary and suspended in a container filled with water. A downward force is momentarily applied to the object. Assuming the water density remains constant and ignoring the effects of fluid friction, what will happen to the object immediately after this downward force is removed?

- ✔ ☒ A. The object will sink at a constant velocity. [34%]
- ☐ B. The object's downward velocity will increase with time. [3%]
- ☐ C. The object will quickly stop sinking. [34%]
- ☐ D. The object will sink before returning to its initial position. [27%]

Correct

35% answered correctly

Explanation:



An object that is **suspended** (submerged and stationary) is neither sinking nor rising because it has the same density as the surrounding fluid. Hence, the upward directed **buoyant force** F_B equals the downward directed weight of the object W :

$$F_{\text{net}} = W - F_B = 0$$

Therefore, the **net force** on the object is initially **zero**, and the object's initial velocity is zero.

When a downward **force** F is **applied** to the object, the net force on the object is downward. The object will sink, and its **velocity** in the downward direction **increases** with time (ie, the object accelerates) as described by **Newton's second law** ($F = ma$):

$$F_{\text{net}} = F + W - F_B = ma$$

Immediately after the applied force is removed, the **net force** acting on the object returns to **zero** (ignoring fluid friction and assuming constant water density). In the absence of a net force, an object at rest (velocity = 0 m/s) will stay at rest, and an object in motion at a given velocity will stay in motion at the same velocity (**Newton's first law**). Therefore, the object will continue to sink at a **constant velocity**.

(Choice B) Immediately after the force is removed, the velocity of the object will remain constant, not increase with time.

(Choice C) An object that experiences a net force of zero will remain at its current velocity; it does not imply that the object will stop moving.

(Choice D) A net force pointing upward would be required to return the object to its initial position. After the applied force is removed, the net force returns to zero because neither F_B nor W depend on depth.

Educational objective:

When a mass experiences a net force, the mass will accelerate in the direction of the force ($F = ma$). When no net force acts on an object, its velocity remains unchanged. Therefore, an object will accelerate to a velocity when a force is applied and will continue at the same velocity after the force is removed.

Time spent:0 sec

Subject:
Physics

QID:401076

System:
Fluids